



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE20CS351 - Cloud Computing

Total Marks : 100.0

1.a. Explain any 4 key design considerations that is used to build a cloud-ready application for private or public clouds (6M). Also, explain how Elasticity plays an important role in cloud computing? (2M) (8.0 Marks)

1.b. Compare and contrast private and public clouds clearly describing the advantages and disadvantages provided by each one of them. (6.0 Marks)

1.c. Explain briefly Bit Level parallelism, Instruction Level parallelism and Task Level parallelism. (6.0 Marks)

2.a. What is the difference between a hosted hypervisor and a bare metal hypervisor. Give an example of each (4M)
List any four differences between a VM and a Container (4M) (8.0 Marks)

2.b. Explain kubernetes architecture with a neat diagram clearly showing all the key components of both master and worker nodes. (6.0 Marks)

2.c. What is the difference between hot migration and cold migration? Explain pre-copy and post-copy techniques of hot migration. (6.0 Marks)

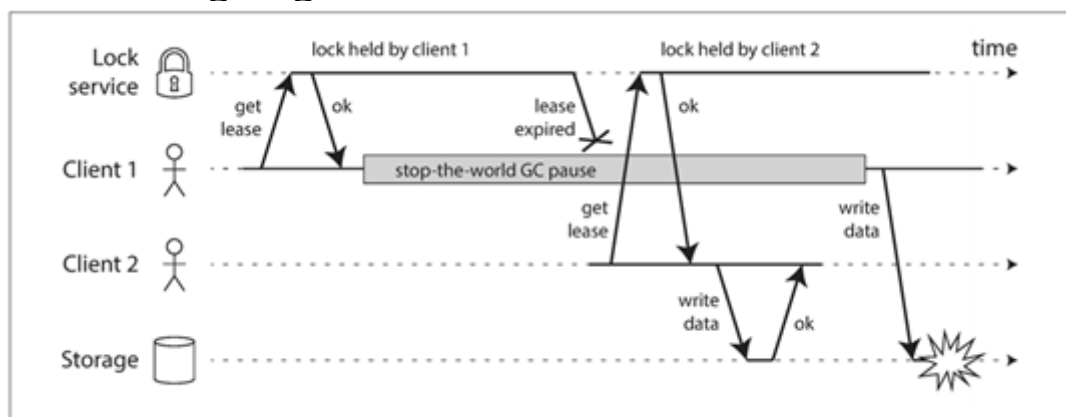
3.a. Explain Gluster file system architecture with a neat diagram.
(8.0 Marks)

3.b. Discuss 3 important properties of the CAP theorem and some of its practical implications while choosing a database for an application based on CAP theorem.
(6.0 Marks)

3.c. What is the purpose of rebalancing of partitions? (2M) Explain Dynamic partitioning and Partitioning proportionally to the nodes (4M)
(6.0 Marks)

4.a. Explain Ring Election Algorithm with neat sketches. Clearly state the worst-case scenario and messages required in worst case scenario. (6M)
What are the changes made in Modified Ring Election Algorithm to address the problem in Ring Election Algorithm? (2M) (8.0 Marks)

4.b. What is the problem with the implementation of a distributed lock in the following diagram? (2M)



What is a "fault-tolerant" system? Name at least two types of failures. (2M)
If a service was unavailable for 60 minutes in 75 hours due to 6 failures, compute the MTBF and MTTR of that service? (2M) (6.0 Marks)

4.c. What is Zookeeper? How does it work? Name at least two contexts where zookeeper services may be used. (6.0 Marks)

5.a. Explain the following terms used in Cloud Security:

Domain in Keystone

Defense in Depth

Honeypot Design Pattern

Network Pattern

(8.0 Marks)

5.b. What is a DoS attack? Explain with appropriate sketches, how is DoS different from DDoS? Distinguish EDoS from the above. (1M+3M+2M)
(6.0 Marks)

5.c. What is a reverse proxy and what are its benefits? How is it different from a forward proxy? Provide a few applications where both are used.
(2M + 2M + 2M) (6.0 Marks)